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Dual basket attachment

Abstract

An implement has an implement frame and a wheel for supporting the implement frame on a ground surface for movement in a direction of travel. The implement also includes a first basket coupled to the implement frame by a first arm and configured to work a portion of the ground surface, and a second basket coupled to the implement frame by a second arm and positioned rearward of the first basket relative to the direction of travel. The second basket is configured to work the portion of the ground surface after the first basket. A hydraulic cylinder is operatively coupled to the first arm and/or the second arm and configured to lift and lower the respective basket with respect to the ground surface to and from a raised position for travel. The hydraulic cylinder is further configured to regulate a pressure of the respective basket against the portion of the ground surface.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. Non-Provisional patent application Ser. No. 16/007,365, filed, Jun. 13, 2018, which claims priority to U.S. Provisional Application No. 62/527,689, filed Jun. 30, 2017, the entire contents of which are herein incorporated by reference.

BACKGROUND

(1) The present disclosure relates to an attachment for an implement such as a tillage tool.

SUMMARY

(2) In one aspect, the disclosure provides an implement including an implement frame, and a wheel for supporting the implement frame on a ground surface for movement in a direction of travel. The implement also includes a first basket coupled to the implement frame by a first arm. The first basket is configured to work a portion of the ground surface. The implement also includes a second basket coupled to the implement frame by a second arm and positioned rearward of the first basket relative to the direction of travel. The second basket is configured to work the portion of the ground surface after the first basket has worked the portion of the ground surface. The implement also includes a hydraulic cylinder operatively coupled to the first arm and/or the second arm and configured to lift and lower the respective basket with respect to the ground surface to and from a raised position for travel. The hydraulic cylinder is further configured to regulate a pressure of the respective basket against the portion of the ground surface.

(3) In another aspect, the disclosure provides an implement including an implement frame, and a wheel for supporting the implement frame on a ground surface for movement in a direction of travel. The implement also includes a first basket coupled to the implement frame by a first arm. The first basket is configured to work a portion of the ground surface. The implement also includes a second basket coupled to the implement frame by a second arm and positioned rearward of the first basket relative to the direction of travel. The second basket is configured to work the portion of the ground surface after the first basket has worked the portion of the ground surface. The implement also includes a hydraulic cylinder operatively coupled to the first arm and/or the second arm and configured to regulate a pressure of the respective basket against the portion of the ground surface.

(4) In yet another aspect, the disclosure provides an implement including an implement frame, and a wheel for supporting the implement frame on a ground surface for movement in a direction of travel. The implement also includes a first basket coupled to the implement frame by a first arm. The first basket is configured to work a portion of the ground surface. The implement also includes a second basket coupled to the implement frame by a second arm and positioned rearward of the first basket relative to the direction of travel. The second basket is configured to work the portion of the ground surface after the first basket has worked the portion of the ground surface. The implement also includes a hydraulic cylinder operatively coupled to the first arm and/or the second arm and configured to lift and lower the respective basket between a raised position for travel and a lowered position for operation.

(5) In yet another aspect, the disclosure provides an implement including an implement frame, and a wheel for supporting the implement frame on a ground surface for movement in a direction of travel. The implement also includes a first basket coupled to the implement frame by a first arm. The first basket is configured to work a portion of the ground surface. The first basket also includes a second basket coupled to the implement frame by a second arm and positioned rearward of the first basket relative to the direction of travel. The second basket is configured to work the portion of the ground surface after the first basket has worked the portion of the ground surface. The implement also includes a first hydraulic cylinder operatively coupled to the first arm, and a second hydraulic cylinder operatively coupled to the second arm. The first and second hydraulic cylinders are independently operable to lift and lower the respective baskets between a raised position spaced above the ground surface for travel and a lowered position in engagement with the ground surface for operation.

(6) Implements such as tillage tools can be used to break up soil clods, remove weeds, and cover seed. Both shank style and disk blade style tillage tools can create soil profiles in the soil having convolutions of laterally spaced hills and valleys commonly described as shank tracks on shank

machines or disk blade marks on disk style machines. These convolutions of the soil can be undesirable to plant into, so various harrows have been designed and utilized on shank style tillage machines and on disk blade style tillage machines to level these convolutions out prior to planting. In high residue conditions, several of the current harrow solutions in the market today either struggle to do a good job of leveling the soil surface out or they struggle to flow residue evenly while leveling the soil surface. The present application provides a unique solution to level low to high residue seedbeds without creating residue bunching.

(7) In some embodiments, the disclosure provides a vehicle and implement mounted thereto. The vehicle and implement include a first wheel coupled to the vehicle for supporting the vehicle on soil, a second wheel coupled to the implement for supporting the implement on the soil, and a motor for driving the vehicle and guiding the implement on the first and second wheels in a direction of travel. The vehicle and implement further including an implement frame supported by the second wheel, a first basket coupled to the implement frame, the first basket configured to work the soil such that any convolutions in the soil are reoriented to be substantially perpendicular to the direction of travel of the vehicle and implement, and a second basket coupled to the implement frame and positioned rearward of the first basket in the direction of travel, the second basket configured to work the soil such that the convolutions in the soil that are substantially perpendicular to the direction of travel of the vehicle and implement are flattened.

(8) In some embodiments, the disclosure provides an implement having an implement frame, a wheel for supporting the implement frame on soil in a direction of travel, a first basket coupled to the implement frame, and a second basket coupled to the implement frame. The first basket works the soil such that any convolutions in the soil are reoriented to be substantially perpendicular to the direction of travel of the implement. The second basket is positioned rearward of the first basket, and works the soil such that the convolutions in the soil that are substantially perpendicular to the direction of travel of the implement are flattened.

(9) In some embodiments, the disclosure provides a method of flattening convolutions in soil including reorienting convolutions in the soil that are substantially parallel to a direction of travel of an implement such that the reoriented convolutions are substantially perpendicular to the direction of travel of the implement by moving a first basket across the soil, and flattening the reoriented convolutions in the soil by moving a second basket across the soil.

(10) In some embodiments, the disclosure provides a tillage tool including a frame, at least one disc, and at least one wheel to which a dual basket attachment is coupled. The frame can include a main portion, a first lateral portion, and a second lateral portion. The frame can also include more side portions in some embodiments. The dual basket attachment includes a front basket and a rear basket. The front basket operates in a passive orientation such that soil is smoothed rather than thrown. The front basket includes knife blades arranged in straight bars. The rear basket follows the front basket and further levels soil to flatten the soil profile. The rear basket can be a round or flat bar basket.

(11) In some embodiments, the disclosure provides a dual basket attachment having a front basket and a rear basket. The front basket operates in a passive orientation such that soil is smoothed rather than thrown. The front basket includes knife blades arranged in straight bars. The rear basket follows the front basket and further levels soil to flatten the soil profile. The rear basket can be a round or flat bar basket.

(12) In some embodiments, the disclosure provides a method of flattening convolutions in the soil profile by reorienting disc blade hills and valleys that are generally parallel to the direction of travel to knife-edge hills and valleys that are generally perpendicular to the direction of travel. Then, the knife-edge hills and valleys are smoothed such that the soil profile is flattened.

(13) Other aspects of the disclosure will become apparent by consideration of the detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a top view of a tractor having a tillage tool and a dual basket attachment according to some embodiments.
- (2) FIG. 2 is a partial rear view of the tillage tool and the dual basket attachment of FIG. 1.
- (3) FIG. 3 is a partial perspective view of the dual basket attachment of FIG. 2.
- (4) FIG. 4 is a side view of the dual basket attachment of FIG. 3.
- (5) FIG. 5 is a partial perspective view of a tillage tool and a dual basket attachment in an operational position.
- (6) FIG. 6 is a schematic view of a soil profile according to some embodiments.
- (7) FIG. 7 is a schematic view of a soil profile according to some embodiments.
- (8) FIG. 8 is a perspective view of a tractor having a tillage tool and a dual basket attachment according to some embodiments.
- (9) FIG. 9 is a side view of the tillage tool and dual basket attachment of FIG. 8.
- (10) FIG. 10 is a perspective view of the dual basket attachment of FIG. 8.
- (11) FIG. 11 is a rear view of a tillage tool including the dual basket attachment of FIG. 8 in a stowed position according to some embodiments.
- (12) FIG. 12 is a side view of the tillage tool and dual basket attachment of FIG. 11.
- (13) FIG. 13 is a side view of a portion of the tillage tool and dual basket attachment of FIG. 12.
- (14) FIG. 14 is a perspective view of a tillage tool and dual basket attachment according to some embodiments.
- (15) FIG. 15 is a side view of the tillage tool and dual basket attachment of FIG. 14.
- (16) FIG. 16 is a perspective view of the dual basket attachment of FIGS. 14 and 15.
- (17) FIG. 17 is a side view of the portion of the tillage tool and dual basket attachment of FIG. 16.

DETAILED DESCRIPTION

(18) Before any embodiments of the disclosure are explained in detail, it is to be understood that the disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The disclosure is capable of supporting other embodiments and of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. The use of “including,” “comprising,” or “having” and variations thereof herein is meant to encompass the items listed thereafter and equivalents thereof as well as additional items. Unless limited otherwise, the terms “connected,” “coupled,” and “mounted” and variations thereof herein are used broadly and encompass direct and indirect connections, couplings, and mountings, and the terms “connected” and “coupled” and variations thereof are not restricted to physical or mechanical connections or couplings. Also, it is to be understood that phraseology and terminology used herein with reference to device or element orientation (such as, for example, terms like “front,” “back,” “up,” “down,” “top,” “bottom,” and the like) are only used to simplify description of the present disclosure, and do not alone indicate or imply that the device or element referred to must have a particular orientation. In addition, terms such as “first,” “second,” and “third” are used herein and in the appended claims for purposes of description and are not intended to indicate or imply relative importance or significance.

(19) In some embodiments, a dual basket attachment can be used in conjunction with a vehicle such as a tractor and an implement such as a tillage tool. The tillage tool can be a harrow that includes a frame, at least one disc or disc gang, and at least one wheel. The tillage tool can also include at least one of some variation of actuator such as a hydraulic cylinder regulated by a control system.

(20) FIG. 1 illustrates one such possible embodiment. The illustrated embodiment includes a tractor **10** having a motor **11** for driving the tractor **10** and a plurality of tractor wheels **12** for supporting the tractor **10** on ground that is typically soil. A tillage tool **14** is coupled to the tractor **10**. The tillage tool **14** includes a frame **15** having a main portion **15a**, a left-side or first lateral portion **15b**, and a right-side or second lateral portion **15c**. The tillage tool **14** includes rows of disc gangs **16** and rows of wheels **17**. The disc gangs **16** are configured to create convolutions in soil such that the convolutions are oriented substantially parallel to a direction of travel **19**. A dual basket attachment **18** includes a front basket **20**, a rear basket **22**, and a coupling bar **23**. The coupling bar **23** couples the dual basket attachment **18** to the back of the tillage tool **14**. The illustrated front basket **20** and rear basket **22** have a main section **20a** and **22a**, a left-side or first lateral section **20b** and **22b**, and a right-side or second lateral section **20c** and **22c**. The aforementioned sections of the front basket **20** and rear basket **22** are staggered. Other embodiments of the dual basket attachment **18** can include fewer or more sections. For example, the tillage tool **14** may include only one frame portion **15** with associated disc gangs **16** and wheels **17**, and the dual basket attachment **18** may include one front basket **20** and one rear basket **22**.

(21) In the illustrated embodiment of FIG. 1, the tractor **10** is generally in front of the tillage tool **14** and the tillage tool **14** is generally in front of the dual basket attachment **18** oriented in the direction of travel **19**. FIG. 1 also illustrates an example of the placement of each front basket **20** or front basket section relative to each rear basket **22** or rear basket section. Each front basket **20** is generally in front of the associated rear basket **22** such that each front basket **20** traverses the soil before the rear basket **22** during operation.

(22) FIGS. 2 and 3 illustrate one possible embodiment of the dual basket attachment **18** in greater detail. The coupling bar **23** is coupled to the back of the tillage tool **14**. The front basket **20** is coupled to the coupling bar **23** by first arms **24** and the rear basket **22** is coupled to the coupling bar **23** by second arms **26**. The first arms **24** include a first section **24a** and a second section **24b**. The first section **24a** and second section **24b** are connected by a rocker bar **25** as shown in FIGS. 2 and 3. The second arms **26** also include a first section **26a** and a second section **26b**. The first section **26a** and second section **26b** are also connected by a rocker bar **27** as shown in FIGS. 2 and 3. The first arms **24** are operated, respectively, by first hydraulic cylinders **28** and the second arms **26** are operated, respectively, by second hydraulic cylinders **29**. In some embodiments of the dual basket attachment **18**, the hydraulic cylinders **28** and **29** can be controlled by a single hydraulic circuit. Other possible embodiments include each set of hydraulic cylinders **28**, **29** operating separately or any of the hydraulic cylinders **28**, **29** operating individually. The first arms **24** and second arms **26** regulate the pressure at which the respective front basket **20** and rear basket **22** press against the ground while in operation. The hydraulic cylinders **28** and **29** can be operated by a control system within the tractor **10**.

(23) As shown in FIGS. 3 and 4, the illustrated front basket **20** is a knife edge basket and includes six knife blades **30** held together by first end plates **31**. Each knife blade **30** includes a knife blade tip **32** and a knife blade side **34**. The front basket **20** is arranged to operate in a passive orientation such that soil is smoothed rather than thrown. The illustrated front basket **20** rotates in the direction of arrow **35** in response to the movement of the tractor in the direction of travel **19** (see FIG. 1). The passive orientation includes knife blades **30** arranged such that the knife blade sides **34** touch the ground before the knife blade tips **32** penetrate soil when operating in the illustrated direction of rotation **35**. The knife blade tip **32** breaks up soil clods and levels soil while the knife blade side **34** firms soil.

(24) The illustrated rear basket **22** is a pipe basket (round bar basket) having eight pipes (round bars) **36** held together by second end plates **37**. Each round bar **36** further firms and levels soil. Other embodiments can include fewer or more knife blades and round bars. Some embodiments may replace the round bars with flat bars.

(25) FIG. 5 illustrates one possible embodiment of a tillage tool **14** with a dual basket attachment

18 in operation. The illustrated knife blades **30** are straight bars but can also be spiral bars in some embodiments. The illustrated round bars **36** are spiral bars but can also be straight bars in some embodiments. The tillage tool **14** first moves over the soil and creates disc blade hills and valleys that are generally parallel to the illustrated direction of travel of the tractor as indicated by arrow **19** (some of the valleys are indicated with the reference **38** in FIG. 5). The front basket **20** then moves over the soil and reorients the disc blade hills and valleys (such as valleys **38**) into knife-edge hills and valleys (such as valley **40**) that are generally perpendicular to the illustrated direction of travel. The rear basket **22** then moves over the soil and levels the knife-edge hills and valleys (such as from valley **40** to valley **42**), resulting in complete removal of the valleys **38** that are generally parallel to the direction of travel **19** and substantial filling in of the valleys **40** that are generally perpendicular to the direction of travel **19** as shown on the right side of FIG. 5 (see valleys **42**). In some embodiments, flattening refers to a reduction in the height of the hills and depth of the valleys **40** by about fifty percent to valleys **42** but not necessarily to the extent such that the valleys **40** are completely removed from the soil. In some embodiments, the height of the hills and depth of the valleys **40** are only reduced by between about twenty five percent and about fifty percent. In some embodiments, the front and rear baskets **20**, **22** cause an overall reduction in height of the hills of between about sixty six percent and about seventy five percent.

(26) The following tables illustrate two possible examples according to some embodiments of the present invention. Table 1 and FIG. 6 show a first example which includes dry, mellow soil with minimal residue that is worked by a gang angle of four degrees. The soil height after the disk gangs **16** have worked the soil is about three inches with hills **38a** having a height of about three inches above valleys **38b**. The illustrated hills **38a** and valleys **38b** extend generally parallel to the direction of travel **19**. After the front basket **20** travels across the hills **38a** and valleys **38b**, then hills **40a** and valleys **40b** that extend generally perpendicular to the direction of travel **19** are formed. The illustrated hills **40a** have a height of about three inches above the valleys **40b**. After the rear basket **22** travels across the hills **40a** and the valleys **40b**, then hills **42a** and valleys **42b** are formed. The hills **42a** have a height of about one and a half inches above the valleys **42b**. The illustrated hills **42a** and valleys **42b** extend substantially perpendicular to the direction of travel **19**.

(27) Table 2 and FIG. 7 show a second example which includes relatively wet soil with standing corn residue that is worked by a gang angle of twelve degrees. The soil height after the disk gangs **16** have worked the soil is about six inches with hills **38c** having a height of about three inches above valleys **38d**. The illustrated hills **38c** and valleys **38d** extend generally parallel to the direction of travel **19**. After the front basket **20** travels across the hills **38c** and valleys **38d**, then hills **40c** and valleys **40d** that extend generally perpendicular to the direction of travel **19** are formed. The illustrated hills **40c** have a height of about three or four inches above the valleys **40d**. After the rear basket **22** travels across the hills **40c** and the valleys **40d**, then hills **42c** and valleys **42d** are formed. The hills **42c** have a height of about one and a half to two inches above the valleys **42d**. The illustrated hills **42c** and valleys **42d** extend substantially perpendicular to the direction of travel **19**.

(28) TABLE-US-00001
TABLE 1
Example 1
Type of Soil Dry, mellow soil with minimal residue
Gang Angle 4 degree
gang angle Soil Height After Disk Blades 3" hills parallel to the direction of travel
Soil Height After First Basket 3" hills perpendicular to the direction of travel
Soil Height After Second Basket 1.5" hills perpendicular to the direction of travel
Effect of the First Basket Change orientation of the hills
Effect of the Second Basket 50% reduction in height of the hills
Overall Reduction in Height 50% reduction in height of the hills

(29) TABLE-US-00002
TABLE 2
Example 2
Type of Soil Wetter field with standing corn residue
Gang Angle 12 degree
gang angle Soil Height After Disk Blades 6" tall clods/rootballs generally parallel to the direction of travel
Soil Height After First Basket 3"-4" hills perpendicular to the direction of travel
Soil Height After Second Basket 1.5"-2" hills perpendicular to the direction of travel
Effect of the First Basket Change orientation of the hills and 33-50% reduction in height of

the hills Effect of the Second Basket 50% reduction in height of the hills Overall Reduction in Height 66-75% reduction in height of the hills

(30) FIG. 8 illustrates one possible embodiment of a tractor **110** having a motor **111**, a plurality of wheels **112**, a tillage tool **114**, a frame **115**, rows of disc gangs **116**, rows of wheels **117**, and a dual basket attachment **118**. The illustrated frame **115** has a main portion **115a**, an inner left-side or inner first lateral portion **115b**, an inner right-side or inner second lateral portion **115c**, an outer left-side or outer first lateral portion **115d**, and an outer right-side or outer second lateral portion **115e**. The disc gangs **116** are configured to create convolutions in soil such that the convolutions are oriented substantially parallel to a direction of travel **119**. The dual basket attachment **118** is coupled to the back of the illustrated tillage tool **114** and includes pairs of coupling bars **123**, front baskets **120**, and rear baskets **122**. The coupling bars **123** couple the dual basket attachment **118** to the back of the tillage tool **114**. The illustrated front basket **120** and rear basket **122** have a main section **120a**, **122a**, an inner left-side or first inner lateral section **120b**, **122b**, an inner right-side or second inner lateral section **120c**, **122c**, an outer left-side or first outer lateral section **120d**, **122d**, and an outer right-side or second outer lateral section **120e**, **122e**. The aforementioned sections of the front basket **120** and rear basket **122** are staggered. Other embodiments of the dual basket attachment **118** can include fewer or more sections.

(31) FIG. 9 is a side view of the dual basket attachment **118** and the tillage tool **114**. FIG. 10 illustrates the dual basket attachment **118** detached from the tillage tool **114**. As shown in FIGS. 9 and 10, the front baskets **120** are coupled to the coupling bars **123** by respective first arms **124** and the rear baskets **122** are coupled to the coupling bar **123** by second arms **126**. The first arms **124** are operated, respectively, by first hydraulic cylinders **128** and the second arms **126** are operated, respectively, by second hydraulic cylinders **129**.

(32) FIGS. 11-13 illustrate one possible stowed position of the embodiment shown in FIG. 8. To move from a first, operational position to a second, stowed position, the frame **115** and the dual basket attachment **118** are first lifted up and away from the wheels **117** by the hydraulic cylinders **130** as shown in FIG. 12 to prepare for storage. Referring again to FIG. 8, from the operational position the outer first lateral portion **115d** and its corresponding outer first lateral sections **120d**, **122d** and the outer second lateral portion **115e** and its corresponding outer second lateral sections **120e**, **122e** are folded in towards the main portion **115a** by hydraulic cylinders. The inner first lateral portion **115b** and its corresponding inner first lateral sections **120b**, **122b** and the inner second lateral portion **115c** and its corresponding inner second lateral sections **120c**, **122c** also are folded in towards the main portion **115a** by hydraulic cylinders. All parts can be folded simultaneously by activating all of the aforementioned hydraulic cylinders at the same time or the outer parts can fold before the inner parts by activating some of the aforementioned hydraulic cylinders before others. The final stowed position of FIGS. 11-13 shows the main portion **115a** and main sections **120a**, **122a** still generally parallel to the ground while all other portions and sections are now generally perpendicular to the ground, resulting in a compact stowed position.

(33) FIG. 12 illustrates a deployed (or lowered) position of the front basket **120a** and the rear basket **122a** and FIG. 13 illustrates a stowed (or raised) position of the front basket **120a** and the rear basket **122a**. The first hydraulic cylinders **128** and second hydraulic cylinders **129** can remain activated as shown in FIG. 12 such that the first arms **124** and the front basket **120** and the second arms **126** and the rear basket **122** can still be lowered as if still in operation. The first hydraulic cylinders **128** and second hydraulic cylinders **129** can both be actuated to raise the front basket **120a** and the rear basket **122a** as shown in FIG. 13. The stowed position can be beneficial for traveling such that the front basket **120a** and the rear basket **122a** are spaced above the ground surface during traveling.

(34) The stowed position shown in FIGS. 11-13 allows for both easy transportation and for compact storage of the tillage tool **114** and the dual basket attachment **118**. In some embodiments, the inner and outer portions and sections can be stowed in positions that are not generally

perpendicular to the ground including some stowed positions where the inner and outer portions and sections are parallel to the ground and positioned partially above the main portion **115a** and sections **120a**, **122a** and some stowed positions where the inner and outer portions and section are diagonal with respect to the ground.

(35) FIGS. **14-17** illustrate another possible embodiment of a tractor **210** having a motor **211**, a plurality of wheels **212**, and a tillage tool **214** including a frame **215** having a main portion **215a**, a first left-side lateral portion **215b**, a second right-side lateral portion **215c**, rows of ground-engaging tillage tools **216**, rows of wheels **217**, and a dual basket attachment **218**. The ground-engaging tillage tools **216** are configured to create convolutions in soil such that the convolutions are oriented substantially parallel to a direction of travel **219**.

(36) As shown in greater detail in FIGS. **15-17**, the dual basket attachment **218** is coupled to the back of the illustrated tillage tool **214** and includes pairs of coupling bars **223**, a row of tine harrow attachments **225**, front baskets **220**, and rear baskets **222**. The coupling bars **223** couple the dual basket attachment **218** to the back of the tillage tool **214**.

(37) With specific reference to FIG. **16**, the illustrated front baskets **220**, rear baskets **222** and tine harrow attachments **225** have a main section **220a**, **222a**, **225a**, a first left-side lateral section **220b**, **222b**, **225c**, and a second right-side lateral section **220c**, **222c**, **225c**. The aforementioned sections of the front baskets **220**, rear baskets **222**, and tine harrow attachments **225** are generally aligned, but could be staggered in some configurations. Other embodiments of the dual basket attachment **218** can include fewer or more sections.

(38) FIG. **17** illustrates one of the coupling bars **223** which couples one of the front baskets **220**, rear baskets **222** and tine harrow attachments **225** to the vehicle **210**. The illustrated tine harrow attachment **225** is coupled to the coupling bar **223** by a first arm **230**. The illustrated tine harrow attachment **225** is a coil spring and is oriented at an acute angle with respect to the ground surface in the direction of travel **219**. The tine harrow attachment **225** is configured to deflect in the direction of arrow **232** upon contacting one or more clumps of soil. The coil spring configuration of the tine harrow attachment **225** biases the tine harrow attachment **225** against the ground surface during operation. A second spring **234** is coupled to the coupling bar **223** and the first arm **230** to bias the tine harrow attachment **225** downward. There are a plurality of apertures on the first arm **230** to which the second spring **234** can be connected to adjust a downward biasing force of the tine harrow attachment **225**.

(39) The front basket **220** is coupled to the coupling bar **223** by a second arm **236**. The illustrated second arm **236** includes a first section **236a** and a second section **236b** that are connected by a rocker bar **238**. A first hydraulic cylinder **240** is coupled to the first section **236a** and to the coupling bar **223**. The first hydraulic cylinder **240** can raise or lower the front basket **220** with respect to the ground surface. The first hydraulic cylinder **240** can be actuated to adjust the downward force on the front basket **220** during operation as well as to move the front basket **220** to a raised position for travel.

(40) The rear basket **222** is coupled to the coupling bar **223** by a third arm **244**. The illustrated third arm **244** includes a first section **244a** and a second section **244b** that are connected by a rocker bar **246**. A second hydraulic cylinder **248** is coupled to the first section **244a** and to the coupling bar **223**. The second hydraulic cylinder **248** can raise or lower the rear basket **222** with respect to the ground surface. The second hydraulic cylinder **248** can be actuated to adjust the downward force on the rear basket **222** during operation as well as to move the rear basket **222** to a raised position for travel.

(41) In some embodiments of the dual basket attachment **218**, the hydraulic cylinders **240** and **248** can be controlled by a single hydraulic circuit. Other possible embodiments include each set of hydraulic cylinders **240** and **248** operating separately or any of the hydraulic cylinders **240** and **248** operating individually. The hydraulic cylinders **240** and **248** can be operated by a control system

within the tractor **210**.

(42) Various features and advantages of the disclosure are set forth in the following claims.

Claims

1. An implement comprising: an implement frame; a wheel for supporting the implement frame on a ground surface for movement in a direction of travel; a first basket coupled to the implement frame by a first arm, the first basket configured to work a portion of the ground surface; a second basket coupled to the implement frame by a second arm movable independent from the first arm, the second basket and positioned rearward of the first basket relative to the direction of travel, the second basket configured to work the portion of the ground surface after the first basket has worked the portion of the ground surface; and a hydraulic cylinder operatively coupled to the first arm or the second arm and configured to move the respective basket relative to the implement frame to lift and lower the respective basket with respect to the ground surface to and from a raised position for travel, wherein in the raised position the respective basket is spaced above the ground surface, the hydraulic cylinder further configured to regulate a pressure of the respective basket against the portion of the ground surface, wherein the respective basket is configured to be moved to the raised position by actuating the hydraulic cylinder.
2. The implement of claim 1, wherein the hydraulic cylinder is a first hydraulic cylinder operatively coupled to the first arm, the implement further comprising: a second hydraulic cylinder operatively coupled to the second arm and configured to move the second basket relative to the implement frame to lift and lower the second basket with respect to the ground surface to and from a raised position for travel, wherein in the raised position the second basket is spaced above the ground surface, and further configured to regulate the pressure of the second basket against the portion of the ground surface, wherein the first basket is configured to be moved to the raised position of the first basket by actuating the first hydraulic cylinder and the second basket is configured to be moved to the raised position of the second basket by actuating the second hydraulic cylinder.
3. The implement of claim 2, wherein the first and second hydraulic cylinders are controlled by a hydraulic circuit.
4. The implement of claim 2, wherein the first and second hydraulic cylinders are operable individually.
5. The implement of claim 1, further comprising a plurality of tines disposed in front of the first basket and the second basket relative to the direction of travel, wherein the second basket is disposed at a rear distal end relative to the direction of travel.
6. The implement of claim 1, wherein the first basket includes a knife edge basket and the second basket includes a pipe basket, wherein the pipe basket includes at least one round bar extending transverse to the direction of travel and configured to work the portion of the ground surface after the knife edge basket has worked the portion of the ground surface.
7. An implement comprising: an implement frame; a wheel for supporting the implement frame on a ground surface for movement in a direction of travel; a first basket coupled to the implement frame by a first arm, the first basket configured to work a portion of the ground surface; a second basket coupled to the implement frame by a second arm movable independent from the first arm, the second basket and positioned rearward of the first basket relative to the direction of travel, the second basket configured to work the portion of the ground surface after the first basket has worked the portion of the ground surface; and a hydraulic cylinder operatively coupled to the first arm or the second arm and configured to move the respective basket relative to the implement frame to lift and lower the respective basket between a raised position for travel in which the respective basket is spaced above the ground surface, and a lowered position for operation in which the respective basket is in engagement with the ground surface, wherein the implement is operable with one of the first basket and the second basket in the raised position and the other of the first basket and the

second basket in the lowered position.

8. The implement of claim 7, wherein the hydraulic cylinder is a first hydraulic cylinder operatively coupled to the first arm and configured move the first basket relative to the implement frame to lift and lower the first basket between the raised position for travel and the lowered position for operation, the implement further comprising: a second hydraulic cylinder operatively coupled to the second arm and configured move the second basket relative to the implement frame to lift and lower the second basket between the raised position for travel and the lowered position for operation, wherein the first basket is configured to be moved to the raised position of the first basket by actuating the first hydraulic cylinder and the second basket is configured to be moved to the raised position of the second basket by actuating the second hydraulic cylinder.

9. The implement of claim 8, wherein the first and second hydraulic cylinders are controlled by a hydraulic circuit.

10. The implement of claim 8, further comprising a coupling bar operatively disposed between the implement frame and the first and second arms, wherein the first hydraulic cylinder is disposed between the coupling bar and the first arm, and wherein the second hydraulic cylinder is disposed between the coupling bar and the second arm.

11. The implement of claim 7, wherein the respective basket is configured to be spaced above the ground surface in the raised position and configured to be in engagement with the ground surface in the lowered position, and wherein a position of the respective basket is adjusted solely by actuating the hydraulic cylinder.

12. An implement comprising: an implement frame; a wheel for supporting the implement frame on a ground surface for movement in a direction of travel; a first basket coupled to the implement frame by a first arm, the first basket configured to work a portion of the ground surface; a second basket coupled to the implement frame by a second arm and positioned rearward of the first basket relative to the direction of travel, the second basket configured to work the portion of the ground surface after the first basket has worked the portion of the ground surface; a first hydraulic cylinder operatively coupled to the first arm; and a second hydraulic cylinder operatively coupled to the second arm; wherein the first and second hydraulic cylinders are independently operable to move the respective baskets relative to the implement frame to lift and lower the respective baskets between a raised position spaced above the ground surface for travel and a lowered position in engagement with the ground surface for operation, wherein the first basket is configured to be moved to the raised position of the first basket by actuating the first hydraulic cylinder and the second basket is configured to be moved to the raised position of the second basket by actuating the second hydraulic cylinder.

13. The implement of claim 12, wherein the implement is operable with the first basket in the raised position and the second basket in the lowered position.

14. The implement of claim 12, wherein the implement is operable with the first basket in the lowered position and the second basket in the raised position.

15. The implement of claim 12, wherein the implement is operable with the first basket in the raised position and the second basket in the raised position.
